WORKED SOLUTION

STEP II 2005, Question 4

Pure • Inverse Tangent Identities

STEP II 2005, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q4

The key idea

Sums of inverse tangents are handled by taking the tangent of both sides and using $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$. The condition $bc = a^2 + 1$ is exactly what makes the denominator factorise against the numerator. After that the question is about structure: the first result is a rule for *splitting* one inverse tangent into two, and the four-term identity is that rule applied twice, once at each level of a tree. For the numerical part, work backwards to find which splitting produced the four numbers.

Solution

The first identity

Let $A = \arctan\frac{1}{a+b}$ and $B = \arctan\frac{1}{a+c}$. Since $a, b, c > 0$, both $\frac{1}{a+b}$ and $\frac{1}{a+c}$ are positive, so $A, B \in (0, \frac{\pi}{2})$ and $A + B \in (0, \pi)$. Using $bc = a^2 + 1$,

$$\begin{aligned}(a+b)(a+c) - 1 &= a^2 + ab + ac + bc - 1 \\ &= 2a^2 + ab + ac = a(2a + b + c).\end{aligned}$$

Hence, by the addition formula,

$$\begin{aligned}\tan(A+B) &= \frac{\dfrac{1}{a+b} + \dfrac{1}{a+c}}{1 - \dfrac{1}{(a+b)(a+c)}} = \frac{(a+c)+(a+b)}{(a+b)(a+c)-1} \\ &= \frac{2a+b+c}{a(2a+b+c)} = \frac{1}{a},\end{aligned}$$

where we may cancel because $2a + b + c > 0$.

Ask yourself Does $\tan(A+B) = \frac{1}{a}$ really give $A + B = \arctan\frac{1}{a}$?

Not automatically: tan has period π, so we need $A+B$ to lie in $(-\frac{\pi}{2}, \frac{\pi}{2})$, the range of arctan. We know $A+B \in (0,\pi)$. Also $(a+b)(a+c)-1 = a(2a+b+c) > 0$, so $\tan A \tan B < 1$ and the denominator in the addition formula is positive; together with a positive numerator this gives $\tan(A+B) > 0$, which rules out $A+B \in [\frac{\pi}{2}, \pi)$. (Indeed $\tan(A+B)$ is undefined at $\frac{\pi}{2}$ and negative on $(\frac{\pi}{2}, \pi)$.) So $A+B \in (0, \frac{\pi}{2})$, and since $\arctan \frac{1}{a}$ is the unique angle in that interval with tangent $\frac{1}{a}$,

$$\arctan \frac{1}{a+b} + \arctan \frac{1}{a+c} = \arctan \frac{1}{a}. \qquad \blacksquare$$

The four-term identity

Apply the first identity three times; each time all three numbers involved are positive.

- With $a = p+q$, $b = s$, $c = t$: the condition is $st = (p+q)^2 + 1$, which is given, so

$$\arctan \frac{1}{p+q+s} + \arctan \frac{1}{p+q+t} = \arctan \frac{1}{p+q}.$$

- With $a = p+r$, $b = u$, $c = v$: the condition $uv = (p+r)^2 + 1$ is given, so

$$\arctan \frac{1}{p+r+u} + \arctan \frac{1}{p+r+v} = \arctan \frac{1}{p+r}.$$

- With $a = p$, $b = q$, $c = r$: the condition $qr = p^2 + 1$ is given, so

$$\arctan \frac{1}{p+q} + \arctan \frac{1}{p+r} = \arctan \frac{1}{p}.$$

Adding the first two lines and using the third, the sum of the four inverse tangents is $\arctan \frac{1}{p}$, as required. $\blacksquare$

The numerical identity

We need $p = 7$, so $qr = 50$. Rather than guess q and r, work back from the four numbers. If 13 and 21 are the pair $a+b$, $a+c$ from one use of the first identity, then $b = 13-a$, $c = 21-a$ and $bc = a^2+1$ gives

$$(13-a)(21-a) = a^2+1 \iff 273 - 34a = 1 \iff a = 8.$$

Similarly, if 82 and 187 are a pair, then

$$(82-a)(187-a) = a^2+1 \iff 15334 - 269a = 1 \iff a = 57.$$

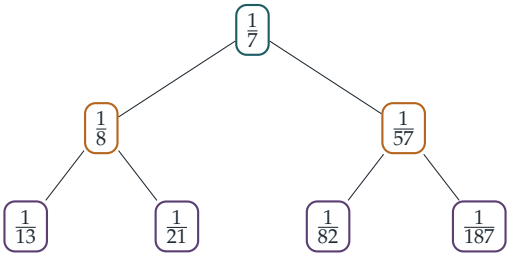
So try $p+q = 8$ and $p+r = 57$, that is $q = 1$, $r = 50$; and then $s = 13-8 = 5$, $t = 21-8 = 13$, $u = 82-57 = 25$, $v = 187-57 = 130$. All seven numbers are positive, and the three conditions hold:

$$qr = 50 = 7^2+1, \qquad st = 65 = 8^2+1, \qquad uv = 3250 = 57^2+1.$$

With these values $p+q+s=13$, $p+q+t=21$, $p+r+u=82$ and $p+r+v=187$, so the four-term identity gives

$$\arctan\frac{1}{13}+\arctan\frac{1}{21}+\arctan\frac{1}{82}+\arctan\frac{1}{187}=\arctan\frac{1}{7}. \qquad \blacksquare$$

The tree below shows how $\arctan\frac{1}{7}$ splits.



Each node's inverse tangent is the sum of its children's: $1\cdot 50=7^2+1$, $5\cdot 13=8^2+1$, $25\cdot 130=57^2+1$.

Checks

- Numerically, both sides of the final identity are $0.1418970546\ldots$.
- Exactly, with fractions: $\tan(A+B)$ for $\frac{1}{13},\frac{1}{21}$ is $\frac{34}{272}=\frac{1}{8}$; for $\frac{1}{82},\frac{1}{187}$ it is $\frac{269}{15333}=\frac{1}{57}$; and for $\frac{1}{8},\frac{1}{57}$ it is $\frac{65}{455}=\frac{1}{7}$.
- A special case of the first identity: $a=1$, $b=1$, $c=2$ gives the famous $\arctan\frac{1}{2}+\arctan\frac{1}{3}=\arctan 1=\frac{\pi}{4}$.

Common slips

- Concluding $A+B=\arctan\frac{1}{a}$ from $\tan(A+B)=\frac{1}{a}$ without checking that $A+B$ lies in $(-\frac{\pi}{2},\frac{\pi}{2})$.
- Searching blindly through factorisations of 50, 65 and 3250; working backwards from the pair 13, 21 finds $a=8$ in one line.

Beyond STEP

Identities like these are behind Machin-type formulae such as $\frac{\pi}{4}=4\arctan\frac{1}{5}-\arctan\frac{1}{239}$, which were used for centuries to compute π by hand, because $\arctan x$ has a rapidly converging series when x is small. The cleanest way to see them is through complex numbers: $\arctan\frac{1}{n}$ is the argument of $n+\mathrm{i}$, so adding inverse tangents corresponds to multiplying Gaussian integers, and the condition $bc=a^2+1$ is a factorisation of $a^2+1=(a+\mathrm{i})(a-\mathrm{i})$ in disguise. Needham's *Visual Complex Analysis*

develops this geometric view of multiplication, and Davenport's *The Higher Arithmetic* covers sums of two squares.

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